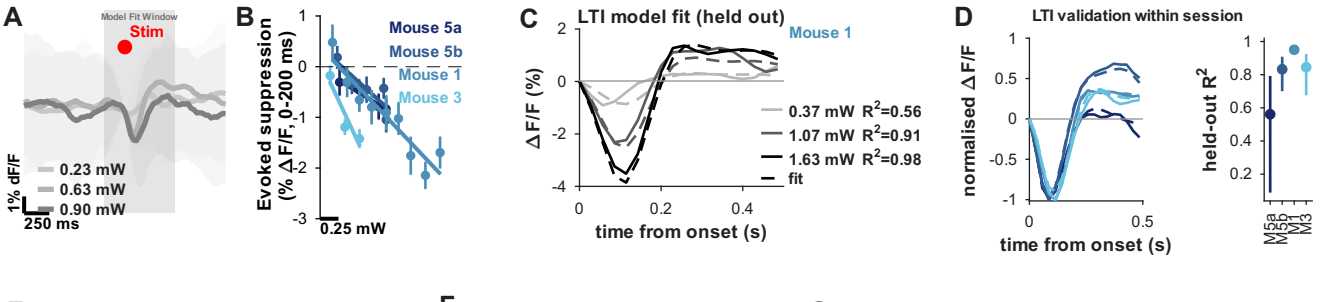

A
Model Fit Window
Stim
0.23 mW
0.63 mW
0.90 mW
1% dF/F
250 ms
B
Evoked suppression (% ΔF/F, 0-200 ms)
Mouse 5a
Mouse 5b
Mouse 1
Mouse 3
0.25 mW
C
LTI model fit (held out)
Mouse 1
ΔF/F (%)
0.37 mW R²=0.56
1.07 mW R²=0.91
1.63 mW R²=0.98
fit
time from onset (s)
D
LTI validation within session
normalised ΔF/F
time from onset (s)
held-out R²
M5a
M5b
M1
M3
E
model from session
applied to session
R² (free gain)
M5a 0.98 0.39 0.72 0.77
M5b 0.12 1.00 0.85 0.73
M1 0.57 0.84 1.00 0.97
M3 0.71 0.68 0.95 0.99
F
Timescales
dip
rebound
0.113 s
τ (s)
G
State dependence of stimulation response
Trial-to-trial variability (× session typical)
Stim
No stim
Q1 Q2 Q3 Q4
Motion
Rel δ
Abs δ power